Malek Dinari

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Summary

Driven computer science engineering student and machine learning enthusiast dedicated to mastering the intricacies of AI and Computer Vision. With hands-on experience in the development of innovative solutions through creativity and analytical thinking, I am committed to continuous growth and learning in the field. Eager to apply my skills in ML, MLOps, or DevOps, I am currently seeking a short-term summer internship (1 month) in July–August 2025 to validate an earlier academic internship requirement. Open to research, software engineering, or machine learning positions.

Education

Bachelor's Degree in Computer Science - ENSI SEPTEMBER 2021-PRESENT

Specialization: Data Science and Computer Vision Status: Graduated (awaiting official diploma issuance)

Related Coursework: Object-oriented programming, Data Structures, Databases, Statistics, Computer Networks, Distributed Systems, Applied Mathematics, Computational Complexity, Quantization, Image Processing, Artificial Intelligence.

preparatory institute of engineering studies of Nabeul (IPEIN) 2019-2021

Mathematics Physics (MP): ranked 193/1779

Baccalaureate Mathematics 2018/2019

Graduated with 14.08/20

Projects

Machine Learning/DL: Bone Marrow Smear Segmentation - ENSI GitHub Link

4-month design and development project:

- Built an AI pipeline to classify and segment bone marrow cells, providing biologists with a diagnostic tool to aid in disease monitoring.
- Fine-tuned MobileNet for initial image segmentation, followed by inference with YOLOv8 to detect multiple cell types within smears, enabling comprehensive cell identification in a single workflow.
- Deployed an interactive frontend using Gradio in a Colab Notebook for accessibility and ease of use in research or clinical settings.
- This tool is tailored for biologists to facilitate bone marrow analysis and improve diagnostic accuracy through automated image analysis.

Mobile Development: WeeFizz Mobile Size Recommendation App - Welyne GitHub Link

3-month summer internship project:

- Developed the mobile app for WeeFizz, a size recommendation tool, as part of a project with Welyne, an IT services startup. The app was built using React Native (community CLI, bare workflow) and integrates seamlessly with the existing WebApp.
- Implemented core functionalities, including QR code scanning (or manual URL entry), user parameter collection (e.g., height, body type), and photo capture for personalized size suggestions.
- Designed to support Prestashop users, the app enhances the shopping experience by offering precise clothing size recommendations based on individual body measurements.
- Gained mobile development experience in the context of practical e-commerce solutions while developing a deeper interest in roles focused on ML and DevOps.

Projects

ML/NLP & AI: Mini-GPT Model - Self-Directed GitHub Link

Mini-project (in progress):

- Implemented a foundational GPT-like model from scratch using Shakespeare's complete works as training data, with a focus on understanding core concepts in generative pre-trained transformers and attention mechanisms.
- Designed and trained an initial Bigram Language Model as a baseline for autoregressive text generation, while exploring scalable architectures that enhance linguistic coherence and depth.
- Expanding the model to incorporate trigram and n-gram components, with a future goal of integrating a multilayer perceptron (MLP) for increased capacity and scalability in handling more complex language structures.
- This project aims to reinforce knowledge in natural language processing (NLP), particularly in attention-based architectures, while providing a stepping stone towards developing larger models or advanced text generation pipelines.

Computer Vision: 4D Facial Expression Recognition - ENSI (Ongoing) GitHub Link

Part of a university research project supervised by Dr. Faouzi GHORBEL:

- Developing algorithms for facial expression recognition using 3D representations (curved 2D surfaces, point clouds, or meshes).
- Implementing geodesic potential descriptors for level curve extraction and geodesic distance computation (Dijkstra's algorithm).
- Exploring advanced Computer Vision techniques such as SIFT detectors and Iterative Closest Point (ICP) for mesh matching.
- Aimed at creating robust 3D or 4D models for improved biometric and expression classification.
- Tools & Skills: Python, Matplotlib, Plotly, PCA, Linear Algebra, 3D Visualization.

ML: Salary Calculator Interface Github Link

- Trained and deployed an AI model via Streamlit to calculate salaries based on the 2020 Stack Overflow Developer Survey.

Selected Computer Vision & ML Notebooks GitHub Link

Ongoing personal project:

- Created a structured set of Jupyter notebooks covering fundamental techniques and algorithms in computer vision and machine learning, organized into four primary modules:
  - **Pattern Recognition:** Implementation of essential techniques such as Hu's 7 moments, Ghorbel complex moments invariance, and Jan Flusser's rotation invariants.
  - **Mathematical Morphology:** Exploration of morphological operations, including geodesic dilation reconstruction, erosion-dilation, and advanced Laplacian and top-hat transformations.
  - **Image Processing and Restoration:** Covering key concepts like DFT/IDFT, Fast Fourier Transform (FFT), Gaussian and motion blurring, kernel-based filtering, and Wiener filters, with additional introductory work in deep learning and optimization.
  - **3D Mesh Processing & Face Expression Analysis:** Focused on facial expression recognition and 3D mesh representation, including level curve analysis using geodesic distances and methods for representing curves on facial meshes.
- Aims to provide a comprehensive reference for foundational and advanced topics in computer vision and image processing, with ongoing updates and new notebooks planned or to be added.

Projects

PFE: Non-Invasive Blood Pressure Estimation via rPPG from Arm Video Captures GitHub Link

End Of Studies project, 4 months Developed a novel signal processing and machine learning pipeline for estimating systolic and diastolic blood pressure using remote photoplethysmography (rPPG) from video captures. Tackled challenges like motion robustness and signal integrity. The project was conducted in collaboration with CRISTAL GRIFT Laboratory and Caire AI (AI healthcare startup based in germany). Results are confidential.

Full-Stack Application: Provider Management System GitHub Link

- Developed a complete CRUD application using Spring Boot for the backend and Angular for the frontend, hosted on PHPMyAdmin for database management.
- Implemented user-friendly features, such as a provider list display and a form for adding new providers, showcasing strong full-stack expertise.
- Gained experience in Spring Suite 4, REST APIs, and Angular component-based design for dynamic web single-page applications.

NLP/C Programming: Semantic and syntactic analyzer Github Link

- Created an app that performs semantic and syntactic analysis for sentences and expressions.

Relevant Skills

Programming languages & Frameworks	C/C++/Java-Intermediate, JavaScript-Intermediate, TypeScript-Beginner,SQL-Intermediate, Python-Intermediate, PyTorch/Tensorflow/Keras-Intermediate, React Native-Intermediate, NodeJS-Beginner, Flutter-Beginner, Angular-Intermediate, Spring Boot-Intermediate
Technologies & Tools	Kaggle/Colab/Jupyter Notebook, DevOps tools(Git/GitHub, Docker), Gradio, Postman (REST APIs testing), Streamlit
Language Skills	English-Fluent, Arabic-Native, French-Proficient, Spanish-A1:Beginner, German-A1:Beginner

Interests

Machine Learning	Reading & Researching	Music
Mathematics	Data Science	Chess and Strategy Games

Last updated: November 10, 2025